

# Prateek H Kumbar

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## PROFESSIONAL SUMMARY

Electronics & Telecommunication Engineering undergraduate with practical expertise in **Embedded Systems, IoT Firmware Development, Power Profiling**, and **Hardware-Software Co-Design**. Proficient in Embedded C, Python, microcontrollers (ESP32, ATmega328P), serial protocols (I2C, SPI, UART), and 4G LTE telemetry. Experienced in breadboarding, circuit simulation, bench testing with multimeters/oscilloscopes, and writing low-latency interrupt-driven routines (ISRs). Seeking an **Embedded Systems / IoT / Electronics Engineering Internship**.

## EDUCATION

### Bangalore Institute of Technology

Bachelor of Engineering in Electronics & Telecommunication Engineering

CGPA: **7.64 / 10.0**

Bengaluru, India

2023 – 2027

**Relevant Coursework:** Embedded Systems, IoT Architecture, Microcontrollers, Analog & Digital Circuits, Power Electronics, Signal Processing, Communication Systems, Circuit Simulation.

### BLDEA's Sri B M Patil PU College

Pre-University Course (PCMC) – **94.3%** | Class X (Secondary Education) – **90.88%**

Vijayapur, India

2021 – 2023

## TECHNICAL SKILLS

- **Programming Languages:** Embedded C, C, C++, Python, SQL, OOP Concepts, Data Structures basics.
- **Microcontrollers & Architectures:** ESP32, ESP32-CAM, Arduino Uno (ATmega328P), ARM Cortex-M fundamentals, FreeRTOS (basics).
- **Protocols & Interfacing:** I2C, SPI, UART, GPIO, ADC, PWM, Timers & ISR (Interrupt Service Routines), 4G LTE, Wi-Fi (802.11 b/g/n), MQTT, TCP/IP, HTTP.
- **Hardware & Instrumentation:** Power Profiling, Circuit Prototyping, Breadboarding, Soldering, LM2596 Buck Converters, Digital Multimeter, Oscilloscope basics, Logic Analyzers, Sensor Interfacing (MPU-6050, Neo-6M GPS, DHT11, INA219, Flame Sensors).
- **EDA, Simulation & Dev Tools:** Proteus, LTspice, MATLAB, PSpice, Keil uVision, Arduino IDE, ESP-IDF, Git, GitHub, LaTeX.
- **Core Competencies:** Firmware Debugging, Hardware-Software Troubleshooting, Real-Time Telemetry, Technical Documentation.

## KEY TECHNICAL PROJECTS

### Adaptive Voltage Scaling & Power Profiling for IoT Nodes | ESP32, Embedded C

01/2026 – Present

- Engineered an empirical power-profiling testbed to characterize silicon-level energy consumption under dynamic clock and voltage states, verifying quadratic power reduction ( $P_{\text{dynamic}} \propto V_{DD}^2 \cdot f$ ).
- Programmed firmware to dynamically toggle MCU clock rates across 4 tiers (40 MHz to 240 MHz) while stepping supply voltage from 3.3V to 3.0V, systematically profiling current draw and latency across I2C, SPI, UART, and Wi-Fi bursts.
- Formulated a multi-dimensional telemetry dataset using Python (NumPy/Pandas) to analyze voltage-frequency trade-offs and train future TinyML workload-aware power management models.

### IoT-Enabled Wireless Car Black Box System | ATmega328P, 4G LTE, Embedded C++

08/2025 – 12/2025

- Developed a standalone automotive safety black box for automated vehicular collision and fire hazard telemetry with zero reliance on smartphone or external network pairing.
- Implemented crash-detection logic computing Euclidean norm acceleration vectors ( $\sqrt{a_x^2 + a_y^2 + a_z^2}$ ) from 6-axis MEMS data, triggering an interrupt service routine (ISR) at  $2.5g$  ( $25 \text{ m/s}^2$ ) threshold in  $< 100 \text{ ms}$ .
- Integrated Neo-6M GPS (locking coordinates in  $< 42 \text{ s}$  with CEP  $< 10 \text{ m}$ ) and automated Hayes AT command sequences over 4G LTE, dispatching emergency SMS alerts with Google Maps links in under 8 seconds.

## CERTIFICATIONS

- **Python Essentials 1** – Cisco Networking Academy
- **SQL and Relational Databases 101** – IBM Cognitive Class
- **Data Analytics Job Simulation** – Deloitte (Forage): Data Analysis & Forensic Technology
- **Introduction to Cybersecurity** – Cisco Networking Academy
- **Getting Started with Artificial Intelligence** – IBM SkillsBuild

## LEADERSHIP & CO-CURRICULAR ACTIVITIES

- **President, Telecommunication Engineering Students Association (TESA):** Headed student chapter operations, coordinated cross-functional technical committees, and directed logistics for **INCEPTRA 2026** (National-level Ideathon and Technical Poster Presentation).